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Wave propagation

TFE4130 Exercise Lab

Department of Electronic systems

For this exercise you must pair up and work together. For the first task you will calculate some microstrip lengths and have it checked by teaching assistants. You perform the other tasks together in the lab. This exercise is approved when you both have completed the lab.

1. Microstrip matching network

In assignment 7 you calculated the lengths and impedances of two matching networks. We will create microstrip lines of these matching networks and we need to calculate the physical dimensions first. To find the correct dimensions we need to know the parameters of the PCB board and its dielectric material. We are using FR4 with parameters:

Parameter	Value	Units	Explanation
ϵ_r	4.46		Relative permittivity
$tan\delta$	0.019		Dielectric loss tangent
h	35	μm	Conductor thickness
d	1.47	mm	Substrate thickness
μ_r	1	H/m	Relative permeability
σ_c	$5.96 \cdot 10^7$	S/m	Conductor conductivity

Table 1: FR4 parameters

Your antenna is resonant at 2.25 GHz. Here are some helpful equations that can also be found in *Microwave engineering: theory and techniques* by David Pozar, chapter 3.8, John wiley & sons, 2021.

$$Z_0 = \begin{cases} \frac{60}{\sqrt{\epsilon_e}} ln\left(\frac{8d}{W} + \frac{W}{4d}\right) & for\ W/d \leq 1 \\ \frac{120\pi}{\sqrt{\epsilon_e}(W/d + 1.393 + 0.667 \cdot ln(W/d + 1.444))} & for\ W/d \geq 1 \end{cases} \tag{1}$$

$$\frac{W}{d} = \begin{cases} \frac{8e^A}{e^{2A} - 2} & for\ W/d < 2 \\ \frac{2}{\pi}\left[B - 1 - ln(2B-1) + \frac{\epsilon_r - 1}{2\epsilon_r}\left(ln(B-1) + 0.39 - \frac{0.61}{\epsilon_r}\right)\right] & for\ W/d > 2 \end{cases} \tag{2}$$

$$A = \frac{Z_0}{60}\sqrt{\frac{\epsilon_r + 1}{2}} + \frac{\epsilon_r - 1}{\epsilon_r + 1}\left(0.23 + \frac{0.11}{\epsilon_r}\right) \tag{3}$$

$$B = \frac{377\pi}{2Z_0\sqrt{\epsilon_r}} \tag{4}$$

$$\text{Effective permittivity:} \quad \epsilon_e = \frac{\epsilon_r + 1}{2} + \frac{\epsilon_r - 1}{2} \cdot \frac{1}{\sqrt{1 + 12d/W}} \tag{5}$$

$$\text{Phase velocity:} \quad v_p = \frac{c}{\sqrt{\epsilon_e}} \quad , \quad \text{Propagation constant:} \quad \beta = k_0\sqrt{\epsilon_e} \tag{6}$$

$$\text{Wavenumber:} \quad k_0 = \frac{2\pi}{\lambda} \tag{7}$$

a) You must now calculate the physical length and width of the 50Ω line, as well as the length and width of the quarter wave transformer you found in exercise 7. The answer should be in mm.

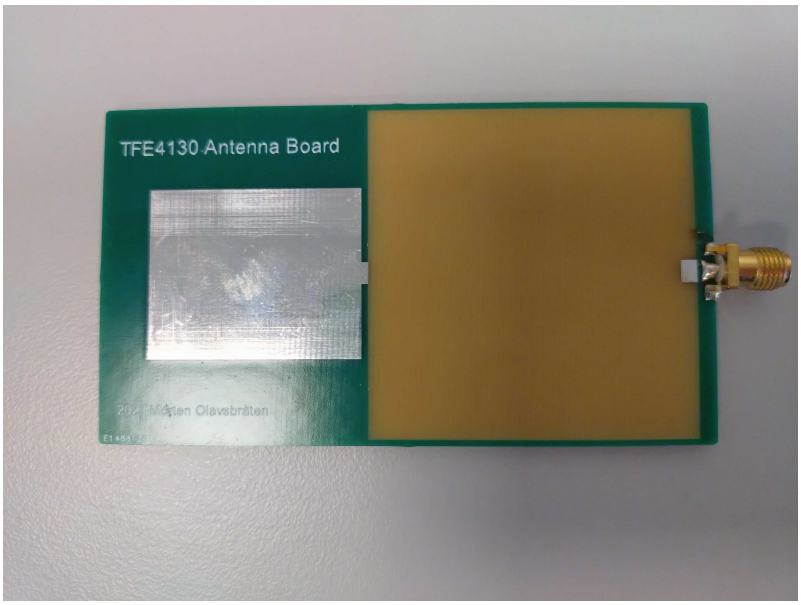
*Hint: The width of the 50Ω line should be *in the ballpark* of 3 mm.

b) Now calculate the physical length of the line and stub matching components you found in exercise 7. The answer should be in mm.

Now you are done with the prep work. The rest of the tasks will be done in the lab.

2. Making the matching network

We are going to work with copper tape that is 3.1 mm thick. This should close enough to the width of the theoretical 50Ω line you calculated in the previous task. If you got values far away from 3.1 mm, you *can* try to implement this design and see what happens.



The board shown in the picture has the antenna on the left side. The antenna is not connected to anything at the moment. You must connect it to the SMA connector using copper tape and doing your best to realize your matching network design in the space between.

- You may draw on the board using a pencil.
- Removing tape can be done with a scalpel, but be careful not to make cuts and scratches in the substrate.

- You must use your nail or something else suitable to press the tape down hard. It is important that the connection between the tape and the board is good and even along the entire tape length.
- After pressing the tape down, let it sit for a minute to let the glue settle a bit.
- Measure the S-parameters using a Vector Network Analyzer (VNA). What do you see?
- Hold your hand over the antenna without touching it. What do you see?
- Repeat the process with your second matching circuit. Which one has better performance?

The assignment is passed when you have made and tested both matching networks and showed the results to a lab assistant.